

Jacquard Machine — Control Electronics

Information Request · everything needed to continue the project

This document lists **every question and every photograph** still needed. It is written to be used with the boards in front of you, or carried around the factory.

Machine	Julong GDH122SA — dead, dismantled
Boards in hand	3 — HMI, Main (JITIBAN-4), Racking (SWG-Yarn-A-02)
Selector board	Never recovered from the machine head
Interconnection record	None — boards removed before any cabinet photo
Goal	Drop-in replacement controller
Documented so far	49 photographs, 3 boards, ~40 ICs identified

How this document is organised

PART	WHAT IT COVERS	WHERE
1	How to photograph — the two mistakes that already cost us	bench
2	The four questions that outrank everything else	ask
3–5	The three boards — marked photos showing what to shoot	bench
6	Power supplies	factory
7	Factory visit checklist — with answer lines	factory
8	Quick reference — the five things that matter most	—

Most of this needs no factory visit

Parts 3–5 can all be done today with the three boards you already have, a camera and a multimeter. Even if the factory visit produces nothing, that work still moves the project forward.

Safety — read before powering anything

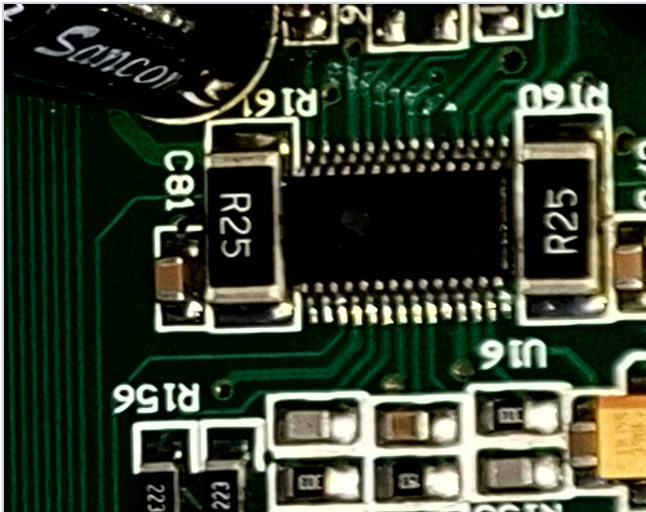
The HMI board carries a **CCFL backlight inverter** marked **DANGER HIGH VOLTAGE** . It makes several hundred volts AC from a low-voltage input and is live whenever the HMI is powered — including from a bench supply. Bulk capacitors hold charge after disconnection.

On the factory floor: if a machine is powered, do not open cabinets or reach into a head. Ask maintenance to isolate it first.

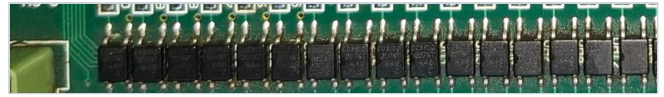
1 · How to photograph

The existing 49 photographs are good. Every failure in them is one of two things, and both are one small change away from being fixed.

Problem 1 — glare hides the chip marking



X Unusable. Main board stepper driver U6 . The package is perfectly sharp — the laser marking simply is not catching any light. All six stepper drivers failed this way.



✓ Readable. Same board, different angle — cosmo 357NT K46 C legible on every package.

Fix: light the board from an **angle**, never head-on. **Turn the flash off.** Tilt the board slowly until the marking catches the light, then shoot. Chip markings are laser-etched, not printed — they only appear at the right angle.

Problem 2 — silkscreen runs off the edge of the frame

The Racking board's year code and both screw-terminal pin labels were lost this way. **Fix:** include about 20 mm of board beyond whatever you are shooting.

Three more worth adopting

- **Get the reference designator in the frame** with every chip shot — the U6 , U28 etc. printed next to it. Half the value of a part number is knowing which position it came from. We currently have photos whose markings are readable but whose board location is unknown.
- **Shoot the solder side too** wherever a connector body hides its own silkscreen.
- **Shoot connectors square-on**, not at an angle — otherwise the pin order cannot be counted from the photo.

File naming

Name by position first, then what it is:

main_U6_stepper-driver.jpg · hmi_J15_power-inlet.jpg · racking_S1_dip-switch.jpg

Take far more photos than this document asks for

You may not get a second chance at the factory. Photographs cost nothing. A missing photograph has already permanently cost this project the entire record of how the machine was wired.

2 · The four questions that outrank everything else

Answer these first. They decide whether the rest of the project is worth doing, and you may be able to answer them from the doorway.

Q1 — Does the dead machine still exist anywhere?

Frame, cabinet, harness, head — any part of it. Scrapyard, back of the workshop, outside under a sheet. This single answer unlocks or permanently closes 12 other items.

Where exactly:

Q2 — Are there OTHER machines of the same model at the factory? ★

Probably the most valuable question in this document. A working sister machine gives you, in one visit: a live reference, a wired cabinet to photograph, a needle bed, and a possible donor selector board.

How many: _____ Model: _____
_____ Running / idle / scrapped: _____

Same controller boards inside? (open one and compare):

Q3 — A replacement controller for *which* machine?

The machine these boards came from is gone, so there is currently no host for a "drop-in" replacement to drop into. The answer changes the whole project.

- ☐ Other machines still in service — how many: _____ ☐ Rebuild this machine
☐ A product to sell to others ☐ Learning / documentation only
-

Q4 — Why was it dismantled? What failed first?

Ask the maintenance person, not the manager. "What stopped working first?" If one board failed, that board's replacement jumps to the front of the queue.

Which board was suspected: _____ Anything replaced before it died: _____

Why Q2 matters so much

When the boards were pulled from the cabinet, no photographs were taken of the cables. That record is gone permanently — we no longer know which supply fed which board, or even **which board drove the machine head**.

Photographing a **sister machine's wired cabinet** recovers all of that by proxy. It is the single most valuable thing that could come out of a factory visit, after a selector board itself.

3 · HMI board — operator panel

Two-board stack: a carrier board plus a plug-in CPU module marked `GOLDEN COREBOARD_V1.20`. The only board in the system with a processor.



HMI carrier board with the CPU module fitted.

#	WHAT IT IS	WHAT IS NEEDED
1	CPU module — S3C2440AL processor,	★ Lift the module off and photograph its UNDERSIDE . The top side has only three chips; the boot flash is not among them, so it must be underneath. That chip holds the

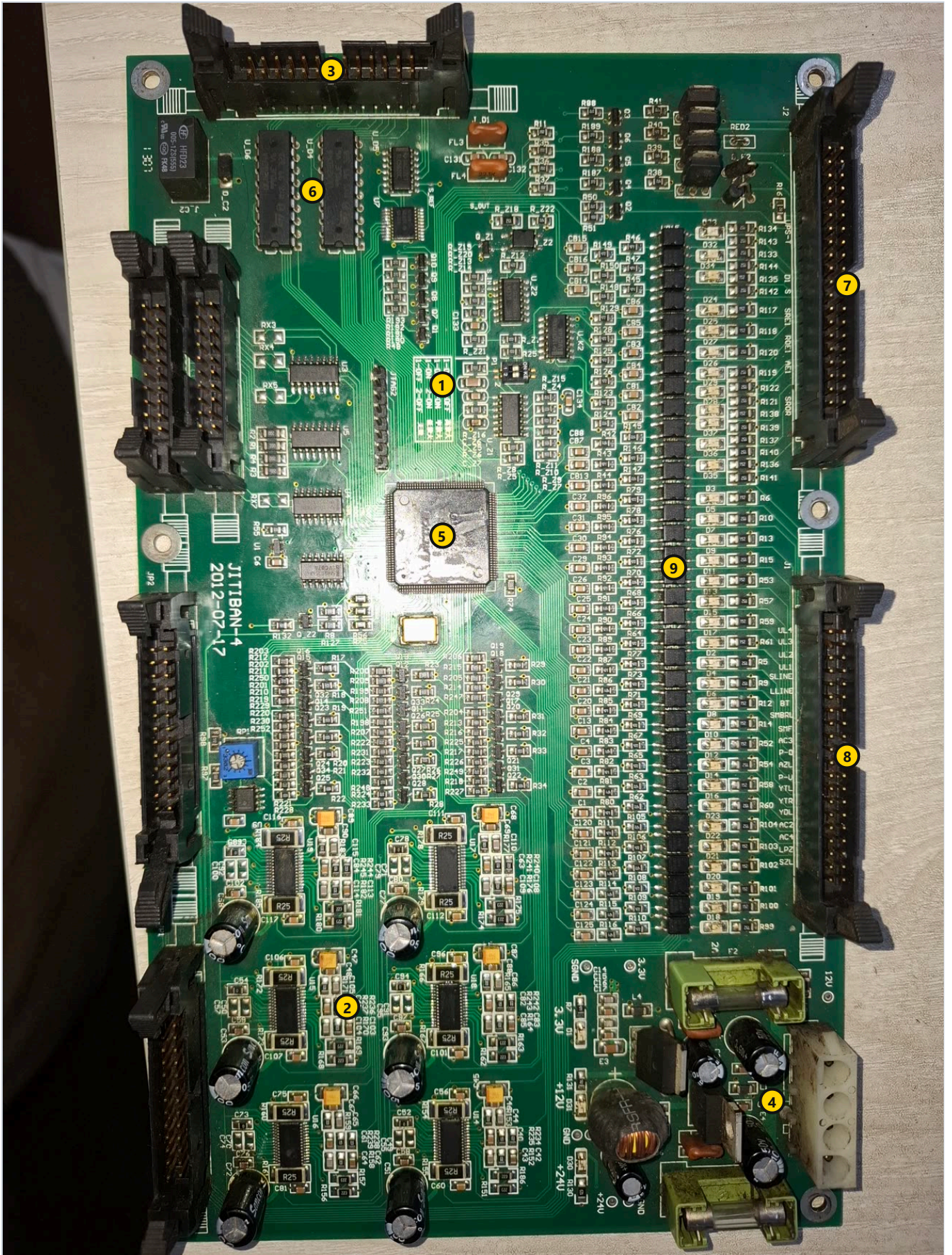
	SDRAM, SRAM	firmware — it is the biggest single unknown we can still recover. Also photograph the empty sockets it plugs into.
2	SW1 — 4-way DIP switch	Record the positions before touching it , then photograph.
3	J15 — DC power inlet	Photograph the inlet and the track running from it through D3 to the big capacitor CE0 and regulator U20 — including the voltage printed on the side of CE0 . The input voltage is printed nowhere on this board and the regulator is a linear type: too much voltage destroys it instantly.
4	Three DB9 sockets — J9, J8, J13	Photograph each square-on with its pin-1 mark. One of these is a serial console; the other two carry the machine bus. We do not yet know which is which.
5	Bus test points — two rows of plated pads	Sharp, square-on photograph. These are the intended probe points for the machine bus and the basis of the whole protocol analysis.
6	JTAG header	Photograph with the surrounding printing and the pin-1 mark.
7	U_D7 — marked FU120N	Closer shot, angled light. The part number is still unidentified.
8	CCFL backlight inverter	High voltage. Photograph only — do not probe while powered.

Also for this board

Every fuse body (F1 , F2) so the ratings can be read · regulators U20 , U21 , U22 with markings legible · both sides of the board, flat and square-on, as a clean re-shoot.

4 · Main board — JITIBAN-4

The machine I/O board: six stepper channels, ~36–38 isolated sensor inputs, 16 output drivers, one safety relay. No processor — everything is done by the FPGA.



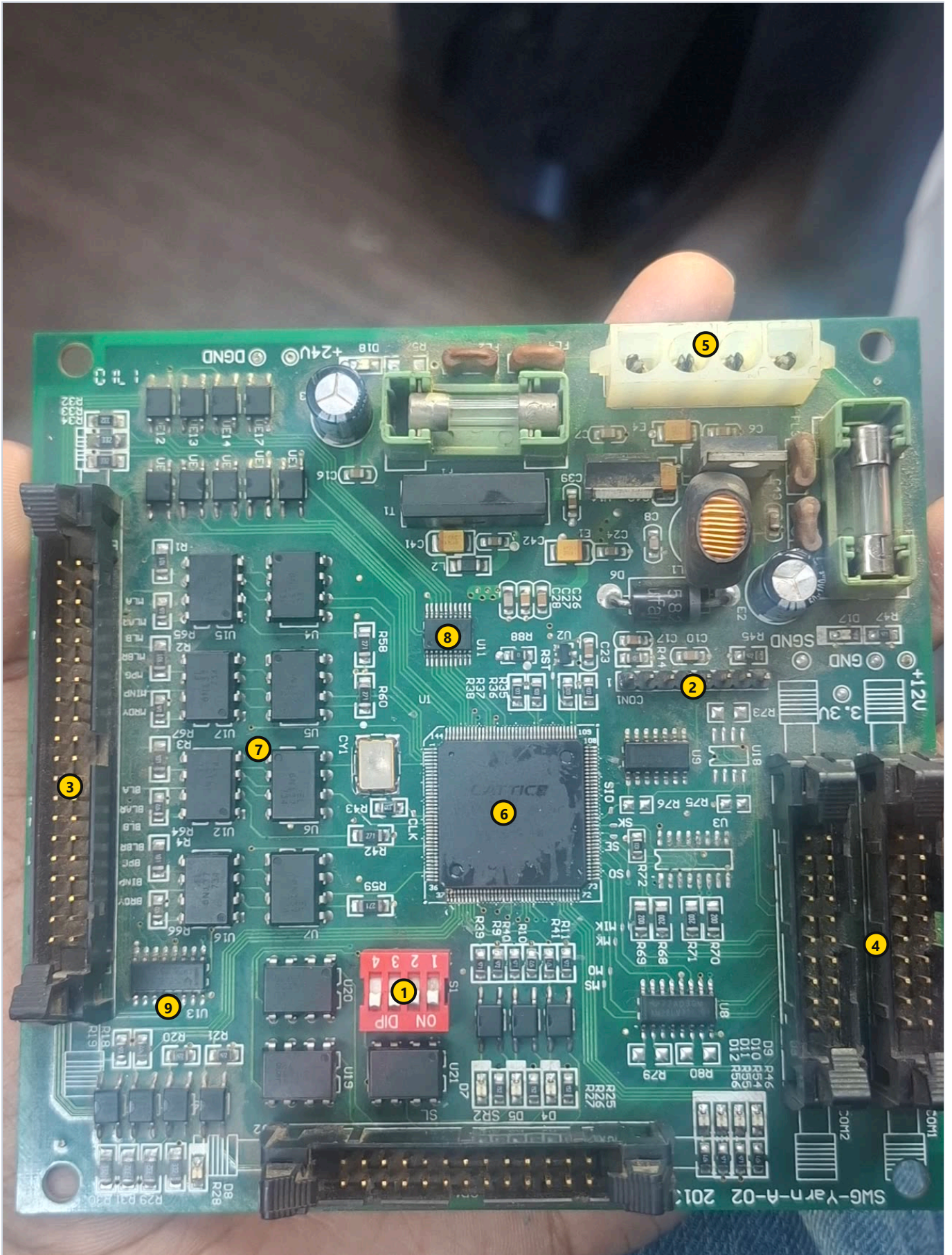
Main board, component side.

#	WHAT IT IS	WHAT IS NEEDED
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1	P1 — 2-way jumper with a printed table	★ Close, square-on photograph of the printed table. It lists four machine modes. The English part is already readable; the Chinese words beside each row are too small in the current photo. Also record the jumper position as found.
2	Six stepper motor drivers	★ One close photograph of each, with angled light and its U-number in frame. Every one is currently unreadable because of glare. These set the motor current and step resolution a replacement must match.
3	40-pin ribbon connector, top edge	Square-on, with any printing around it. We need to work out whether this is the connector that went to the machine head.
4	Power inlet — 4-pin screw terminal, 2 fuses	Photograph the terminal, and the underside of the board beneath it — the terminal body hides its own pin labels. Both fuse bodies so the ratings can be read.
5	U28 FPGA and the J1AG2 programming header	Photograph the header with its pin-1 mark, and the printing around the FPGA.
6	Two 18-pin chips + relay	Nearby chips U3, U5, U7, U17–U19, U_Z1, U_K2, U21, U22 are unidentified — one photo each, designator in frame.
7	J2 connector	Both square-on with pin-1 marks visible, so the pin order can be counted from the photo.
8	J1 connector	
9	Long row of ~36–38 small optocouplers	Already identified — no photo needed. This is the isolated sensor input bank, one per machine sensor.

5 · Racking board — SWG-Yarn-A-02

Satellite node for the racking axis and yarn carriers. Talks to two AC servo drives. Heavily isolated — eight fast optocouplers. No processor.



Racking board, component side.

#	WHAT IT IS	WHAT IS NEEDED
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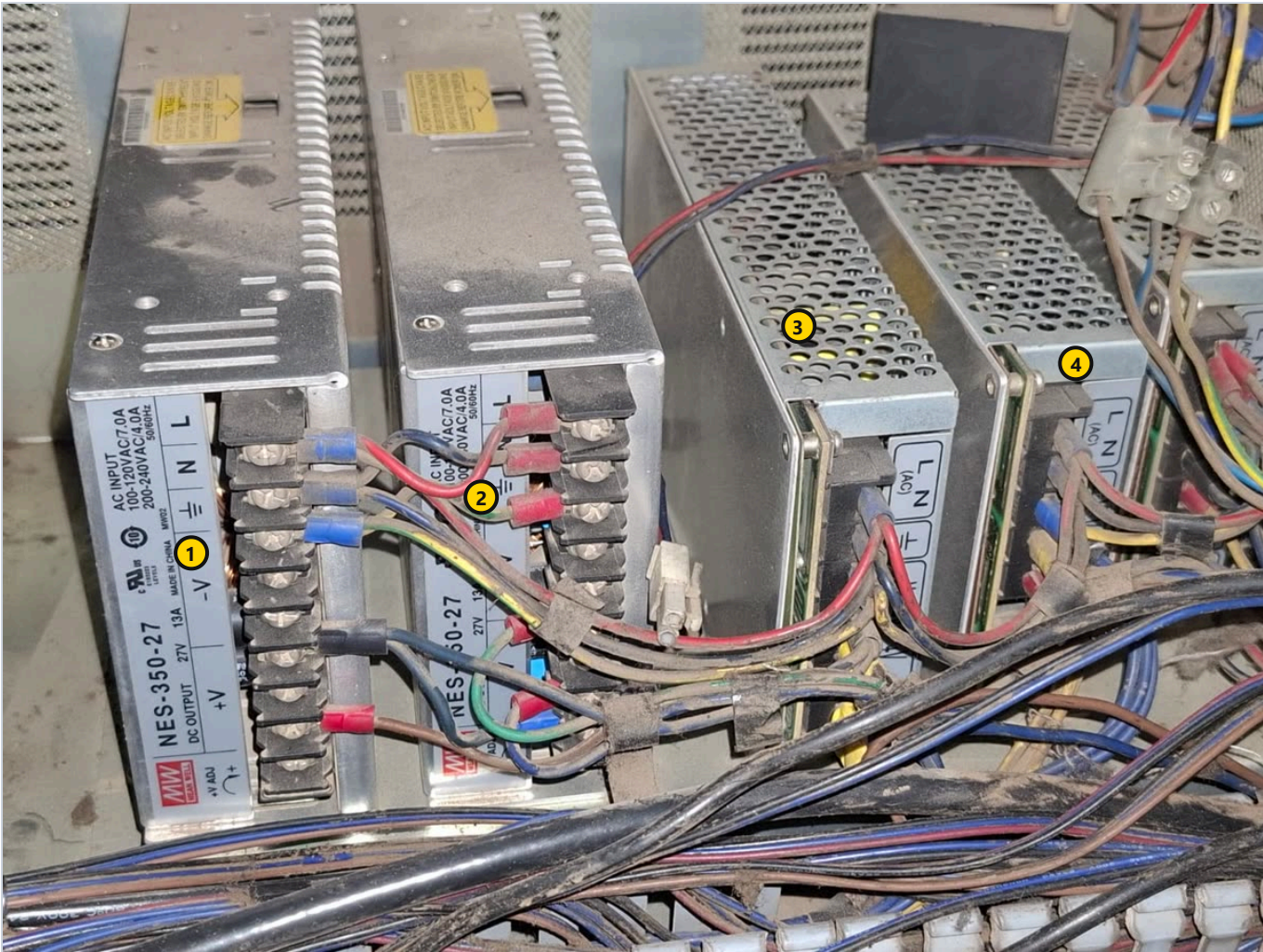
1	S1 — 4-way DIP switch	★ Record all four positions before touching it, then photograph. This was the board's address on the machine bus. With the wiring record gone, this is one of very few surviving clues to how the machine was set up.
2	CON1 — 10-pin programming header	Photograph with pin-1 mark and surrounding printing.
3	34-pin servo connector	Square-on with the printed signal names visible. This is the connector that went to the two AC servo drives.
4	COM1 / COM2	Both square-on with pin-1 marks. These carried the machine bus in and out.
5	Power inlet — 4-pin terminal + fuses	Photograph the terminal, the underside of the board beneath it, and both fuse bodies so the ratings can be read.
6	Lattice FPGA	Photograph the printed labels along its edge, sharp.
7	Eight fast optocouplers	Already identified — no photo needed.
8	U11 — small 16-pin chip	Unidentified. One close photo each, angled light, designator in frame.
9	U13	

For all three boards

Every connector, square-on, with its **pin-1 mark** visible — the little triangle, square pad or printed "1". That is roughly 20 photographs and it is the foundation everything else is built on. Also a clean re-shoot of **both sides of each board**, flat and square-on.

6 · Power supplies

There is no custom power-supply board — the machine used bought-in units. This is the only photograph of them, and two of the four are unidentified.



The supply rack as found in the cabinet.

#	UNIT	STATUS
1	Mean Well NES-350-27	27 V, 13 A, 350 W — label fully readable. Nothing needed.
2	Mean Well NES-350-27	Same. Nothing needed.
3	Unknown	Nameplate needed. These are almost certainly the source of the 12 V supply that the boards need and the 27 V units cannot provide. Until they are identified the machine's total power budget is unknown. Also photograph their output terminals with the wires still attached, and the 115 V / 230 V selector switch position on every unit.
4	Unknown	

An important finding — 27 V, not 24 V

The boards have +24V printed on them, but the supplies feeding that rail are set to 27 V. This looks deliberate: a higher voltage makes the needle selector solenoids switch faster, and allows for voltage lost along the cable to the moving head. The 50 V capacitors on the Main board support this.

So: bench-test the boards at 27 V, not 24 V. And any replacement supply must be adjustable up to 27 V — the LRS-350-24 in the project folder is a 24 V unit and would need turning up.

7 · Factory visit checklist

Fill in the lines. A recorded "don't know" is worth more than a blank — it tells us the question is still open rather than overlooked.

If the dead machine still exists

Q5 — Is the carriage head still on the machine?

☐ Yes ☐ No ☐ Removed and stored separately — where:

Q6 — Is the selector board still inside the head? ★★

The most important physical item in the whole project. It is the board that drives needle selection. Look for a circuit board with many identical driver chips in rows and a thick flexible cable running back to the cabinet. If it is there — **photograph everything, both sides, every chip** — and ask whether you can take it away.

☐ Yes — photographed ☐ Head is empty ☐ Could not open the head

Q7 — Is the cabinet still wired, with the plugs still in?

If yes — **photograph it before unplugging anything**. Every cable, both ends. This is exactly the record that was lost.

☐ Intact and wired ☐ Partly stripped ☐ Empty

Q8 — Servo drives

How many:_____

Make:

Model (photograph the nameplate):

We believe these are Mitsubishi MR-J series, worked out from the signal names printed on the Racking board. Nobody has ever read an actual label.

Q9 — Machine nameplate

Make:_____

Model:

Serial:

Year:

Voltage:

Needle gauge / bed width:

"Julong GDH122SA" currently rests entirely on word of mouth — nothing on any board carries a make or model. Confirming it matters because sourcing a donor selector board needs the exact model.

Q10 — Is the needle bed still intact, with its selector actuators?

Matters because if needle selection has to be designed from scratch, it cannot be done without a bed to test against.

If a sister machine is running

The highest-value part of any factory visit. **Ask permission before opening anything on a working machine.**

Q11 — Are its controller boards the same as ours?

HMI board: _____ Main board: _____

Racking board: _____

Differences noticed: _____

Q12 — Photograph its cabinet fully wired ★★

This recovers, by proxy, the wiring record we lost. Take all of these:

- ☐ Whole cabinet, doors open, every plug still in
- ☐ Every cable, **both ends**, and where each one goes
- ☐ Which power supply feeds which board, and on which terminal
- ☐ **Which connector on which board goes to the machine head**
- ☐ The thick cable from cabinet to head, both ends
- ☐ Terminal blocks and their numbering

Q13 — Its selector board, inside the head

- ☐ Both sides, whole board ☐ Every chip, close, angled light
- ☐ Every connector ☐ Board name and date printed on it

Q14 — Can a selector board be bought, borrowed or swapped?

- ☐ Spare in stock — how many: _____
- ☐ Can be ordered
- ☐ Can borrow one
- ☐ No

Supplier / price: _____

Q15 — Can you watch one running?

- ☐ Short video of a normal knitting cycle
- ☐ Photos of the HMI screen — main menu, settings, any version number

People and paperwork

Q16 — Who maintained this machine, and can you reach them?

Name: _____ Contact: _____

Often worth more than the hardware. They remember how it was wired; nothing else does. Ask them: what plugged into what? which supply fed which board? what did the sensors on **J1** actually connect to? was there a UPS fitted?

Q17 — Any documentation at all?

- ☐ Operator manual ☐ Service manual ☐ Wiring diagrams
- ☐ Parts list ☐ Purchase records ☐ Chinese-language documents ☐ Nothing

Photograph every page of anything you find — do not pick and choose. Chinese is fine, it can be translated. One wiring diagram would answer more questions than months of tracing the boards by hand.

Q18 — Pattern files or design computer?

The HMI loaded knitting patterns over USB. Any surviving memory stick, or the PC that made the patterns, would show the file format.

☐ USB stick found ☐ Design PC exists ☐ Software name:

Q19 — Supplier or dealer the machine came from

Company: _____ Contact: _____

They may still sell the selector board, or have manuals.

Q20 — Factory electrical supply

Voltage: _____ Phase / frequency: _____

8 · Quick reference

The five things that matter most

#	THING	WHY
1	A selector board — from anywhere: the dead machine's head, a sister machine, a spare, a scrapped machine	The only route to understanding how needles are selected. Without it, that half of the system has to be designed from scratch rather than copied.
2	Photos of a wired cabinet on any machine of this model, before anything is unplugged	Recovers the wiring record that was lost — which supply feeds which board, and which board drives the head.
3	Switch positions on the three boards, recorded before anyone moves them	The last surviving evidence of how this machine was configured. Trivially easy to destroy by accident.
4	Every page of any manual or wiring diagram	One diagram beats months of tracing connections by hand.
5	The maintenance person's phone number	They remember the wiring. Nothing else does.

If you can take only one thing away from the factory

Take a **selector board**. If you can take only one photograph, photograph a **sister machine's cabinet fully wired with every plug still in**.

Before you leave the factory

- ☐ Are Q1, Q2, Q3 and Q6 actually answered?
- ☐ Have you photographed a selector board, if one exists anywhere on site?
- ☐ Have you photographed a wired cabinet, if one exists?
- ☐ Switch positions recorded **before** anything was moved?
- ☐ Photos backed up before leaving?
- ☐ Asked whether you can take a board, a spare, or a manual?

What can be done without the factory

Everything in parts 3, 4 and 5 — the photographs of the three boards you already have. Plus, with a multimeter and about **US\$60** of basic tools: tracing out every connector, checking the boards for shorts, and recording the switch positions.

A **US\$25 Lattice JTAG programmer** is the single highest-value purchase: if the chips are not locked, it may recover the original design directly and answer most of the remaining questions at once.

What happens next

Bring back the answers and photographs and they get folded into the project documentation. Several things currently written off as unrecoverable would reopen — particularly if the machine carcass, a sister machine, or a donor selector board turns up.

Generated from the project repository `Jacquard_Machine_IC`. Full technical detail, including the IC list, protocol analysis and replacement plan, is in the `docs/` folder.